

Jachai (যাচাই): merchant-transaction integrity for Bangla QR

Team DIU Comebackhobena — K.M. Sajidul Haq (Captain), Tanjimul Astak Siam, Mohammed Muntasir Rahman Joy · AI DEV FEST 2026 AI Hackathon (DIU CPC x upay) · Track 05, Merchant & Agent Intelligence, with a Trust & Risk engine · Live: <https://jachai-eta.vercel.app> · Code: <https://github.com/kmsajid044-ship-it/jachai>

1. Problem

A customer "pays" a shop Tk 10,000 by Bangla QR, buys nothing, and walks out with about Tk 9,815 in cash. The shop works as an unlicensed cash-out point: the customer bypasses the Tk 30,000 daily cash-out limit and leaves no trail, and upay loses agent cash-out fees (Tk 13–18.50 per Tk 1,000) while paying interchange on off-us QR. From 1 Oct 2026 Bangla QR merchant fees fall to zero with an incentive on payments up to Tk 2,000, which also invites payment splitting; from 1 Nov 2026 P2P Bangla QR creates the reverse risk of business income disguised as personal transfers. Acquirers are expected to monitor merchants with unusual patterns and stop splitting and cash-outs. A blanket limit on every shop is the blunt answer, and it hurts honest wholesalers, electronics sellers and festival traders most.

Sources: team project brief and domain notes (repo CLAUDE.md, "Domain facts"), drawn from the organisers' challenge material [1]. The primary regulatory documents should be cited directly in any version beyond this hackathon.

2. Idea

Don't limit every shop. For every payment and every shop, answer one question: **real commerce, or a hidden cash-out?** Rank the few shops that need review, explain why in plain English and Bangla, leave every decision to a human analyst, and turn confirmed high-demand cash-out points into licensed upay agents.

3. Implemented solution

All data is synthetic and generated by code (make world): 3,000 shops, 30,000 customers and 600 agents over 120 days (Jul–Oct 2026), including the 1 Oct rule change. Six misuse patterns (round-amount cash desk, remittance drain, turnover burst, limit-bypass rings, incentive splitting, P2P disguise) and eight honest look-alikes (big-ticket sellers, bulk buys, festival and haat-day spikes, new shops) are injected by pluggable functions. True labels stay in hidden columns used only for evaluation; models see only rule votes and sparse, dated, shop-level past cases (7% of shops, biased towards rule-flagged shops, with wrong verdicts).

The system has three scores fused into one band:

1. **Payment score:** LightGBM on soft targets (rule votes plus past cases), isotonic calibration, threshold frozen on validation.
2. **Shop score:** expected-turnover regression by category, area and size (robust z-score) plus an Isolation Forest within each peer group.
3. **Network score:** payer–shop graph (NetworkX); shops linked when one payer passes the daily limit across three or more shops on the same day; Louvain communities and a ring flag.

Fusion is an equal-weight average of component ranks (rules included as one input), giving a low / review / high band with cut-offs frozen on validation. A FastAPI service and a Next.js dashboard serve the alert queue, case view, merchant notice, policy simulator and Trust page; a demo mode reads bundled JSON produced by the real API if the API is unreachable.

4. Key features

- Alert queue labelled "needs review", never a verdict; case view with three scores, top-3 reasons in English and Bangla, 30-day timeline and linked-shop graph.
- Decisions (clear, monitor, educate, convert, restrict, escalate) require a reason and go to an append-only, hash-chained SQLite log; nothing is blocked automatically.
- Polite Bangla merchant notice built from the shop's own reasons, with an appeal form preview.
- Policy simulator: the same month replayed under five policies with sliders for misuse share, fee rate, analyst capacity and limit level.
- Trust page: results vs baselines, evasion and ring results, fairness slices, data card and limitations.

5. AI approach

- **Weak supervision.** Eight labelling functions (one per typology plus two "honest" ones) vote; a weighted vote behind a replaceable interface combines them. Trained on rule votes alone, the model only reproduced the rules,

so sparse dated past cases were added as higher-weight evidence.

- **Point-in-time features.** 43 payment, payer and shop features use only information available at payment time; a cut-off test proves it.
- **Explanations.** Exact TreeSHAP values from LightGBM and exact fusion contributions are mapped to reason codes; every number in a reason is filled in by code, and a missing value raises an error instead of becoming zero.
- **Evaluation protocol.** Split by shop and by time; one pattern (turnover_burst) held out of training entirely; thresholds frozen on validation; a single-feature leakage check (all features below 0.95 strength after the synthetic world was made less easy); the test set accessed once and logged in `reports/test_access_log.md`.

6. Results (synthetic data only)

Validation, three seeds (mean $\pm$ std), shop level. Jachai with rules reached PR-AUC 0.774 ± 0.059 against 0.899 ± 0.033 for rules alone and 0.137 ± 0.019 for a blanket limit. Under evasion (misuse amounts made non-round, system not retrained) Jachai held up better: PR-AUC 0.599 ± 0.024 vs 0.588 ± 0.051 , value recall 0.893 ± 0.013 vs 0.739 ± 0.053 , and limit-bypass ring recall 0.591 vs 0.024 . The agreed success criteria for adding rules to fusion were **not all met**: evasion and rings passed, the normal-world criterion (within 0.02 PR-AUC of rules) did not.

Final test, single access (615 test shops, 71 misuse, matched 70-shop budget):

method	PR-AUC	precision	recall	misuse-value recall	honest shops flagged	held-out pattern recall
Jachai with rules	0.584	0.614	0.606	0.774	27	0.133
Rules only	0.843	0.843	0.831	0.894	11	0.600
Blanket limit	0.486	0.371	0.366	0.549	44	0.700

Honest reading. On this synthetic test, rules were stronger than Jachai, including on the held-out pattern, and Jachai flagged more honest shops than the rules (but fewer than a blanket limit). Caveats: the world and the rules were written from the same typologies, which favours rules (circularity); counts are small (29–43 misuse shops per validation seed, 71 at test); one evasion strategy was tested; no real data was used. Jachai's measured advantage is robustness when misusers adapt and network detection of rings, not normal-world accuracy.

7. Real-life impact

The intended value is prioritisation: limited analyst time goes to the shops with the strongest combined evidence; honest big sellers are not caught by a single city-wide cap; confirmed high-demand cash-out points become agent leads, moving cash-out into a licensed, fee-earning channel. In the fast-profile simulator replay (300 shops, 15 days; a diagnostic, not a forecast) a Tk 100,000 blanket limit restricted 12 honest shops and blocked Tk 2.0 million of genuine sales while stopping 25% of misuse taka; rules and Jachai each stopped about 73% with no honest shop restricted.

8. Responsible AI

No real personal data; random IDs; a test scans for phone- or NID-like values. Hidden truth is never served by the API. The system recommends and a named analyst decides, with a written reason, in an append-only log. Merchant wording is polite and non-accusatory, with an appeal path. Fairness: validation false-alarm rates by area type, shop size and category are shown with sample sizes. All results are labelled synthetic.

9. Limitations and next steps

Synthetic data and circularity limit every claim; small slices make some rates unstable; only one evasion strategy is tested; latency and security are prototype grade (SQLite, in-memory scoring). Next: a pilot on real, consented data with analyst-audited labels; a learned label model; more evasion strategies; recalibration and fairness review on real merchants; production storage and access control.

10. References

[1] AI DEV FEST 2026 AI Hackathon Rulebook, General Rules and DIU CPC x upay Student Project Guideline (organisers' documents). [2] Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree", NeurIPS 2017. [3] Lundberg & Lee, "A Unified Approach to Interpreting Model Predictions" (SHAP), NeurIPS 2017. [4] Liu, Ting & Zhou, "Isolation Forest", ICDM 2008. [5] Blondel et al., "Fast unfolding of communities in large networks" (Louvain), J. Stat. Mech. 2008. [6] Ratner et al., "Snorkel: Rapid Training Data Creation with Weak Supervision", VLDB 2017. Libraries and licences: `docs/third_party.md`. Evidence: `reports/summary.md`.

11. AI usage disclosure

The team used AI coding assistants under each member's own account: **Claude Code (Claude Opus 5.5)** and **Codex (GPT-5)**, for code, tests, documentation and analysis. Every AI-assisted task is logged with date, tool and purpose in `docs/ai_usage.md`; the team reviewed, tested and is responsible for all generated work. No AI tool was used to access restricted contest material.